

hardening: 2-axis LayerNorm $\Rightarrow$ BatchNorm (folds into conv)

FIFO $\times 2$

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dual-path GRU $\Rightarrow$ conv

feat $3 \times T \times 257$

$$[|X|^{0.3}, \Delta_f \varphi / \pi, \text{IFD} / \pi] \rightarrow$$

1 $\times$ 1 Conv
3 $\rightarrow$ 6 ch

DSConv 6 $\rightarrow$ 12
low $\frac{1}{4}$: 2×3 , $s=1$
high $\frac{3}{4}$: 2×5 , $s=(1, 3)$

DSConv 12 $\rightarrow$ 18
low $\frac{1}{4}$: 2×3 , $s=1$
high $\frac{3}{4}$: 2×5 , $s=(1, 3)$

DSConv 18 $\rightarrow$ 24
low $\frac{1}{4}$: 2×3 , $s=1$
high $\frac{3}{4}$: 2×5 , $s=(1, 3)$

**dual-path
conv block**

intra-freq $\parallel$ inter-time

$\times 2$ blocks

(deep: $\times 3$, $+d=16$)

skip

mask head
 $2 \times 2, 1 \times 1, \sigma$

USConv 24 $\rightarrow$ 6
low $\frac{1}{2}$: 1×3
high $\frac{1}{2}$: ConvT 1×3 , $s=3$

USConv 36 $\rightarrow$ 12
low $\frac{1}{2}$: 1×3
high $\frac{1}{2}$: ConvT 1×3 , $s=3$

USConv 48 $\rightarrow$ 18
low $\frac{1}{2}$: 1×3
high $\frac{1}{2}$: ConvT 1×3 , $s=3$

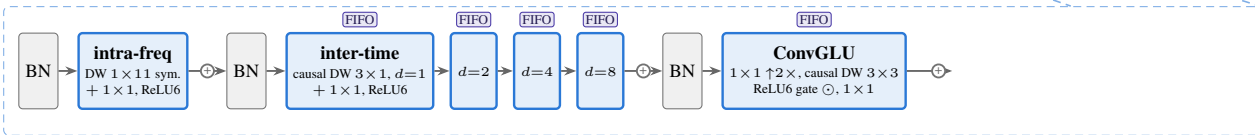
$$\hat{S} = (m_0 + m_1) \cdot |X|^{0.3}$$

$\hat{M}$: $2 \times T \times 257$

PReLU / Mish $\Rightarrow$ ReLU6 (bounded = NPU Clip)

one dual-path conv block (the GRU replacement)

sub-pixel shuffle $\Rightarrow$ ConvTranspose



FIFO causal ring buffer, exposed as state I/O by the streaming export (17 tensors for the $\times 2$ -block model)